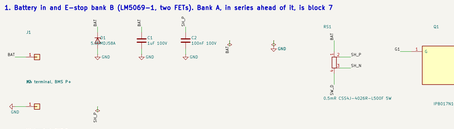
1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7



The bank is the soft start for all the motor capacitance BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it, so a compute crash can't trip it. Two FETs share the heat; the BMS's own FETs never see it. Sized for 100 A continuous.

Start: C4 100 nF, gate 100 Ω, UVL 100→105 V/s, OCL 105→115 V/s. Ramp 0.5 ms at 2.06 A for 0.2 ms, 4.8 A for 0.2 ms). Drive caps unmeasured.

Power limit 652 W at 54.6 V, 480–832 W with tolerance. RPWR = 1.5mΩ + Rs + Rf. = 1.1R mΩ + VDS / ΔIₗ. SNSVSUV2 sq 10. VSNS 0 mV over the 3 mV floor. Timer 4.7 nF, 0.22 ms typ, 0.15–0.39 ms. Retry 1.5 s, 0.1 % duty. Fault 0.5 A, 0.5 ms. Gate 10 V/ms, 1.1 A peak. Inrush 2.1 A, 0.2 ms. Run 0.2–0.3 A, 0.2 ms. Fault 22.7 A at 0.39 ms. Lᵢₙ x. Read the real curve at layout.

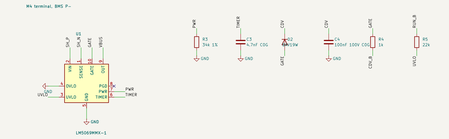
FETs: 0.85 mOhm for the pair, 0.2 W hot at 60 A, 1.3 W at 80 A. The same again in bank A. 10kΩ low sinks the gate at 2 mA. 40uA 170 mA off.

The 2 mA also pulls VBUS down through U1's OUT–GATE zener, against the 5A uA of his internal 1 MOhm SENSE–OUT path, so VBUS reads near 0 V with the banks open.

Reset: 10 kΩ pull-up to VBUS, 100 nF to GND. Low holds the banks open; UVLO held low: opening on overvoltage would shield regen on the bus.

SHDN#=1 latches on a fault/power-on-reset cycle, then reset.

BAT spike: the absorber in block 10 (5 × 47 uF) takes the lead current when a bank opens fast. 5.05MΩ/58A is the backstop, FETs and U1 are 100 V.



Block 10, Absorber: a bank trip into a bus short cuts 350–890 A; tools/absorber_sim.py (pack 54.6 V, no pack yet), worst corners: 0.05 Ohm pulse loop (cell AC resistance), 0.5–1 uH, 81 at 71.2 V, C −20 %, trip 8.8–17.5 us, Three 47 uF caps give BAT 80–92 V; one gave 98–119 V. Measure lead L and pulse R at bring-up. 100k bleeds it. BMS close 0.2–0.3 A, 0.2 ms. Run 0.2–0.3 A, 0.2 ms. Fault 22.7 A at 0.39 ms.

U2 feed: F1 20R, ESD2 in series, then 1 uF, 4 shorted (U) pulls the pack up VCTL 22R holds it to about 1.6 A, so SMAJ58A clamps under 19.2 V (QR gate 20 V) and F1 (50 A breaking) opens. The series diode keeps C5 from back-driving U2 when BAT collapses, and 220R keeps C5R1 inrush far under F1's 50 A breaking option (M3): DNP: If the 16 trips when it closes onto C5R, remove the 5R and fit 2R + the delayed NMOS + 7k 8U; 2R A at close, clamp unchanged (absorber_sim.py).

RUN_A and RUN_B diodes: a 15 µm 2–3 bridge or an E9 fault can't drive RUN. F1 reaches UVLO through BAT46W + 10K (2010) with a 12 V zener on FLT_A; F1' low holds UVLO near 0 V, so U2 can't start. Fault: 0.5 mA, 0.5 ms. Run: 0.2–0.3 A, 0.2 ms.

Converter: F2 20 A on the board near products the pigtail (2? already there), at 9.7–12.5 A; F2 backs up a shorted Q7). The OVD has no reverse protection, so the keyed connector carries polarity. U7 PGD high (good) turns on the 2N7002, which pulls ON/OFF to −Vin through 220R: on (5.45 V at 2 mA), 220R limits a pin 5–11 short to 55 mA; 470k and the 12 V rail make it 0.1 mA. Off: 0 V, 0 mA. Q7 is 100 V.

1 uF on BAT_SNS: SMBJ: a reset held at power-up no longer arms the latch (tools/latch_sim.py, 36 cases). Reset interlock (M1): Q15 shorts RST_SNS until MID is over about 11 V (1.0 nmol, Q15 with Q14's charge, so a reset can't close a bank onto a short). Reset: 10 kΩ pull-up to VBUS, 100 nF to GND. Low holds U1 off; open leaves it to its own logic. Q14 is the latch: it holds MID to about 5 V after a trip, so the U15 MOSFET can't see the full bus. Its UVLO is 5.5 V, so MID below 5 V shuts it off. Q14 is 100 V, 0.5 A, SOT-23. Q15 is 60 V, 0.3 A, SOT-23. Q14 and Q15 are 2N7002s.

It's recommended for LM; BATWOW (NV), ESP_nF on +12V, spare—output shorts and servo back-feed (MID) land in 5 and the compute loads.





12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp, 0.5 W at 54.6 V.

Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

Seal-in latch, regulce shocked at HTT. BD. stays off at power-up even with reset held. reset closes it, a 50 us contact bounce rides through.

Run: momentary, 0.2–0.3 A, 0.2 ms. Fault: 22.7 A at 0.39 ms. Reset: 10 kΩ pull-up to VBUS, 100 nF to GND. Low holds U1 off; open leaves it to its own logic. Q14 is the latch: it holds MID to about 5 V after a trip, so the U15 MOSFET can't see the full bus. Its UVLO is 5.5 V, so MID below 5 V shuts it off. Q14 is 100 V, 0.5 A, SOT-23. Q15 is 60 V, 0.3 A, SOT-23. Q14 and Q15 are 2N7002s.

It's recommendation for LM; BATWOW (NV), ESP_nF on +12V, spare—output shorts and servo back-feed (MID) land in 5 and the compute loads.

Button HB2–BS345, 22 mm mushroom, twist release, 1NC + 1NO, 600 V AC 10 A, control-rated only: NC in this loop, NO to the PI (block 9). RUN also enables bank A (block 7) and SPARE_EN (block 8).

2. Control rail and E-stop latch (no logic IC)

3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into the HS150



Ansyl HS150 108 J on the +X gusset plate, with thermal compound (frame 28777577), leads to J5/J4 on the board's top edge, +X half, twisted. 522 W, 5.7 A while on. MBRS3100 carries the lead and winding inductance at turn-off, so lead length sets EMI; not FET stress. Rated 150 W only on the heatsink; 108 Ω limits it to about 8 W.

Self-powered: VCTL rises as long as the bus or the pack does. If the FET sticks on, the resistor's thermostat opens the loop and the bank drops the bus.

TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. 4ns d/s driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.

Leg stage taken: R1 2 Ohm and recheck 30s TST. 2? A gives about 5 W.

4. Bus capacitors

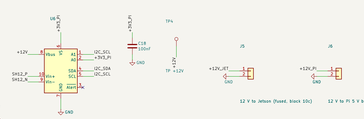
400 uF, 80 V. The drives have their own. Can to 12 × 16.5: land owed, and it only clears under the higher ceiling, away from the fans.



5. 12 V converter: Murata DVE0125V2NBC on the −X front post plate, pigtail J24 (block 10)



12 V converter pigtail: 1–2 +VIN, 3–4 −VIN, 5 ON/OFF (low = on), 6 −PIN, 7–9 −VOUT, 10–12 +VOUT



Ahead of the E-stop: motors off, compute stays up. Soft start in block 5a: the converter's negative-logic ON/OFF is held off until U7's PGD is good (block 10).

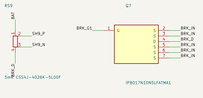
DIP board on the top front rail (frame 28777577), baseplate-down, 4× M3 into its threaded standoffs. 100 mil pin headers, 2.54 mm pitch, 1.78 mm max depth, clear of the board's top edge. −X half, −Vin and −Vout both go to GND; the robot has one ground. P−.

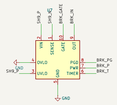
Pigtail about 150 mm, included wire, 18 AWG, 2 wires each way in the cable. 3 each way in the connector. Pin 1 is square on the board and on the housing.

Input UVLO 52–56 V on, 50–54 V off, fixed on the DVE; keep an empty pack under sag, compute can drop before the BMS cuts out.

INA228 addresses, no strap on SDA; I_mrm 0x40, R_mrm 0x41, aux 0x43, I_leg 0x44, R_leg 0x45. SPARE_EN 0x47. watch 0x4C. 12 V 0x4D. MID 0x4F. Powered by the PI's 3.3 V; PI pull-ups on I2C.

5a. Brick soft start (ahead of the E-stop switch)





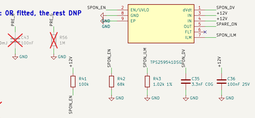


Some controller as the bank, so one part family, −2 retries on its own; a compute fault does not need a BMS cycle. PGD on HN. OVLO low on whenever the pack is.

Fault check: R7 is 1.8 A for up to 5.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the same device as the bank's.

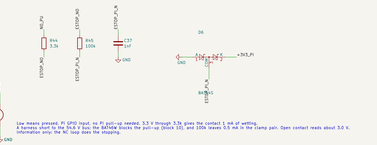
Compute is on wherever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

10b. Absorber,close option (M3): OR fitted, the rest DNP



Unswitched, about 2 A limit. Same part as U9: a short stops in 5 us and leaves compute up.

9. E-stop pressed: button NO contact to the PI

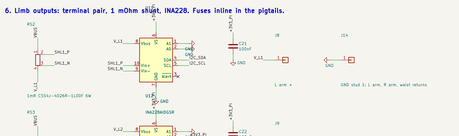


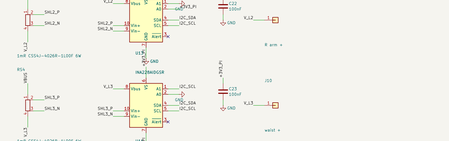
Low means pressed. PI GPIO input, no PI pull-up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.

A harness short to the PI(4.7 kΩ) the BATWOW blocks the pull-up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.

Information only: the NC loop does the stopping.

6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.

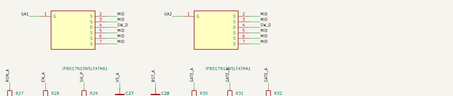






7. E-stop bank A: second series bank, own driver, ahead of bank B







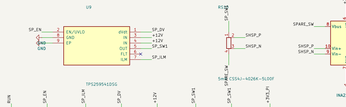
Own driver: a FET usually dies shorted gate to drain, which would pull a shared gate net to the pack and hold the other bank on. RUN is the only shared signal, and each bank takes it through its own diode.

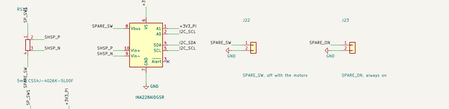
Gate: 10 kΩ to source with no path to GND, so MID can be biased. 47k + 47k. 2x47 k pack nominal, 0.6 mA. 1R mA each. Hot FET leakage 200R uA per bank at 125 °C moves MID by up to about 5 V. DIST reads MID through 33k with a 68 k zener at the pin (85 V max) calibrate the 3.7 k gain.

Sense: 1 mΩ shunt, 1 W, Kelvin. 1 mΩ × 50 A = 50 mV, 2.5 W. INA228 at 0x46 (A1 = SCL, A0 = SDA). Reads V_L1; use the highest limb meter). MID within 2 V of pack: a shorted, MID under 5 V. B shorted, VBUS high: both shorted, or the bus has not bled down yet, about 15 s on board caps, longer with motors.

U17 uses VBUS only. IN+ and IN− on GND, VBUS alone cannot see one shorted bank, which is why MID is read.

8. Spare 12 V outputs

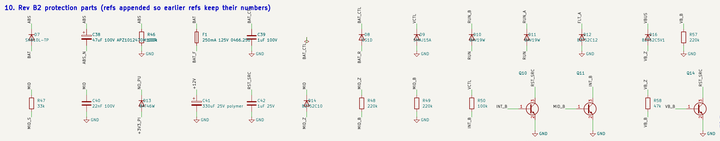




SPARE_SW: RUN enables it, so it drops with the motor bank. The divider keeps 7.7 V of RUN under the 6 V EN limit. Limit 3.87–4.42 A (4870), dV/dt 3.3 V/ms. SPARE_DN: trace 8a. Both spares sit behind R58, so the 12 V meter counts them.

8a. SPARE_ON eFuse

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)



10c. Rev B4 review parts



Rev B4 review: two more absorber caps (three in all), 22 reed 22R and series D18, F17 diode BAT46W (pull-down only), 68 V zener on U17 VBUS, 220R in Q12's drain, 100k on Q14's gate, 100k on Q15's gate, 100k on Q16's gate, 100k on Q17's gate. Q16 and Q17 are 2N7002s. Trace 8a done.

Need the bus. Fuses on the inbuse. PI buck and fan outputs the DVE limits at about 50 W; SMBJ15A on +12V servo back-feed clamp. 15.7 V min breakdown, so it never conducts at 12 V.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR

Author:	marengo
Sheet:	/
File:	marengo_pdu_rev0.kicad_sch
Title: marengo power board rev B	
Size: A2	Date: 2026-11-06
KiCad 10.0.0	Rev: B−wip
	Id: 1/1